

AARON GEORGE VARGHESE

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EDUCATION

Bachelor of Technology in Computer Science Engineering
Junior at Vellore Institute of Technology, Chennai

May 2028
CGPA: 9.27

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, Java, HTML, CSS, React
Frameworks & Libraries: FastAPI, Celery, Redis, React
AI / ML: Retrieval-Augmented Generation, LLM Integration, Agentic Systems, Langgraph
Databases: ChromaDB, PostgreSQL, SQLite
Tools & Concepts: Docker, Tailscale, Supabase, JWT, Data Pipelines

EXPERIENCE

- Sion Healthcare** — Ai Full Stack Intern (March 2026 - Ongoing)
- Building an AI service that converts unstructured natural language input into structured, schema-validated JSON output using LLM inference.
 - Developed multi-stage validation and normalization pipelines for LLM outputs, including domain-specific unit conversion and schema enforcement to ensure data consistency.
 - Developed a secure admin dashboard using Next.js and Supabase.
- Nexrova** — AI / LLM Intern (Oct 2025-Oct 2025)
- Built a FastAPI-based backend for a WhatsApp chatbot using Twilio Webhooks, enabling real-time messaging
 - Designed a message processing pipeline to handle incoming user queries and generate structured, context-aware responses
 - Integrated LLM APIs into messaging workflows, including prompt construction and response parsing
 - Used ngrok to expose local services for external webhook communication, enabling rapid testing and iteration.
- Daira Edtech** — Business Research Intern (Apr 2025 – Jul 2025)
- Conducted in-depth research on market trends, potential clients, and competitor strategies to support data-driven strategic decision-making.
 - Identified new growth opportunities through systematic market analysis, translating findings into actionable recommendations for the leadership team.
 - Collaborated with the business development team by providing structured insights that contributed to pipeline expansion and go-to-market planning.

PROJECTS

- Stride** | Python, Flask, Celery, Redis, React, Docker
- Building and Shipping a multi tenant web app that syncs users Garmin Connect Health data, displays it on a custom dashboard, social features.
 - Designed an async ingestion pipelines (FastAPI + Celery + Redis + PostgreSQL) that pulls the data, upserts it into the database, recovers from auth failures without disconnecting accounts.
 - Implemented JWT auth, encrypted credential storage, rate limiting, per user data isolation to support concurrent accounts safely.
 - Built a social layer with privacy boundaries – exposing performance data while keeping health metric privated
 - Built a React 19 + Recharts dashboard, admin monitoring, a global filter/state architecture.
- Nexus** | Python, LangGraph, FastAPI, ChromaDB, PostgreSQL [GitHub-Nexus](#)
- Built a hybrid business intelligence agent combining Text-to-SQL and RAG to answer natural language queries over structured, unstructured e-commerce data
 - Implemented a Text-to-SQL pipeline that achieved 37% accuracy against the BIRD evaluation framework
 - Designed a multi-node agentic workflow using LangGraph for query routing, SQL generation, retrieval, response synthesis
 - Built a HyDE-based RAG pipeline with ChromaDB for semantic retrieval over unstructured business data
- Echo** | Python, Ollama, Qwen2.5-Coder, Textual, LLMs [GitHub-Echo](#)
- Built a CLI coding agent with local LLM inference for autonomous code generation and debugging
 - Engineered a two stage token aware context compression pipeline to optimize prompt efficiency and manage context window constraints
 - Designed a background-threaded agent loop with event-driven task queuing for non-blocking agent execution and responsive CLI interaction
 - Implemented prompt-engineered tool calling using a custom built JSON parser

ACHIEVEMENTS

- **Hackathons:** Secured 5th Overall Rank in BitWars 2.0, VIT Chennai a competitive technical hackathon.
- **Clubs:** Active member of OSPC (Open Source Programming Club) and CodeYGen.